

COS20019 - Cloud Computing Architecture

Week 10: ACA Module 9 Challenge Lab - Creating a Scalable and Highly Available Environment for the Cafe

Author: Trac Duc Anh Luong - ID: 103488117

Due Date: 12/11/2023

Task 1: Inspecting your environment

In this task, you will evaluate the current state of your lab environment.

5. Explore the lab environment, including how the network is set up.

Tip: You might want to start in the **Amazon VPC console**.

To help you explore, continue to the next section and answer the questions about this lab.

Answering questions about the lab

Answers will be recorded when you choose the blue **Submit** button at the end of the lab.

6. Access the questions in this lab.
 - Choose the Details menu, and choose Show.
 - Choose the **Access the multiple choice questions** link that appears at the bottom of the page.
7. In the page that you loaded, answer the questions:
 - **Question 1:** Which ports are open in the *CafeSG* security group?
 - **Question 2:** Can you connect from the internet to instances in *Public Subnet 1*?
 - **Question 3:** Should an instance in *Private Subnet 1* be able to reach the internet?
 - **Question 4:** Should an instance in *Private Subnet 2* be able to reach the internet?
 - **Question 5:** Can you connect to the *CafeWebAppServer* instance from the internet?
 - **Question 6:** What is the name of the Amazon Machine Image (AMI)?

AWS Academy Cloud Architecting - Module 9 Challenge Lab Questions

View questions in: [English](#)

Question 1: Which ports are open in the *CafeSG* security group?

- ☐ Ports 80 and 443
- ☒ Port 80
- ☐ Ports 80, 443, and 3899
- ☐ Ports 22, 80, and 443

Submit

Question 2: Can you connect from the internet to instances in *Public Subnet 1*?

- ☒ Yes - If the instance has a public IP address, and the security group and network ACL allow it
- ☐ No - The public subnet has no internet gateway
- ☐ No - The public subnet has no NAT gateway configured for it
- ☐ No - The network access control list (network ACL) prevents any inbound traffic from the internet

Submit

Question 3: Should an instance in *Private Subnet 1* be able to reach the internet?

- ☒ Yes
- ☐ No

Submit

Question 4: Should an instance in *Private Subnet 2* be able to reach the internet?

- ☐ Yes
- ☒ No

Submit

Question 5: Can you connect to the *CafeWebAppServer* instance from the internet?

- ☐ Yes
- ☒ No

Submit

Question 6: What is the name of the Amazon Machine Image (AMI)?

- ☐ Amazon Linux
- ☐ WebServerAMI
- ☒ Cafe WebServer Image
- ☐ My Amazing Image

Submit

Task 2: Creating a NAT gateway for the second Availability Zone

To achieve high availability, the architecture must span at least two Availability Zones. However, before you launch Amazon Elastic Compute Cloud (Amazon EC2) instances for your web application servers in the second Availability Zone, you must create a NAT gateway for them. A NAT gateway will allow instances that do not have a public IP address to access the internet.

8. Create a NAT gateway in the *Public Subnet* in the second Availability Zone.
9. Configure the network to send internet-bound traffic from instances in *Private Subnet 2* to the NAT gateway you just created.

The screenshot shows the AWS VPC Console interface for creating a NAT gateway. The breadcrumb navigation is **VPC > NAT gateways > Create NAT gateway**. The page title is **Create NAT gateway** with an **Info** link. A description states: "A highly available, managed Network Address Translation (NAT) service that instances in private subnets can use to connect to services in other VPCs, on-premises networks, or the internet." The **NAT gateway settings** section includes:

- Name - optional**: A text input field containing "Nat Gateway 2".
- Subnet**: A dropdown menu showing "subnet-0135bfa49a0880184 (Public Subnet 2)".
- Connectivity type**: Radio buttons for "Public" (selected) and "Private".
- Elastic IP allocation ID**: A dropdown menu showing "eipalloc-045618ab2ff1a66db".

 An **Allocate Elastic IP** button is located to the right of the Elastic IP dropdown. A green notification bar at the top states: "Elastic IP address 18.233.130.20 (eipalloc-045618ab2ff1a66db) allocated."

The screenshot shows the AWS VPC Console interface for editing routes. The breadcrumb navigation is **VPC > Route tables > rtb-0243f008e63d70947 > Edit routes**. The page title is **Edit routes**. Below the title is a table with the following columns: **Destination**, **Target**, **Status**, and **Propagated**.

Destination	Target	Status	Propagated
10.0.0.0/16	local	Active	No
0.0.0.0/0	NAT Gateway	-	No

 The table shows two routes. The first route has destination 10.0.0.0/16 and target local, with status Active. The second route has destination 0.0.0.0/0 and target NAT Gateway, with status -. There is a search bar and a dropdown for each row. A **Remove** button is next to the second route.

Task 3: Creating a bastion host instance in a public subnet

In this task, you will create a bastion host in a public subnet. In later tasks, you will create an EC2 instance in a private subnet and connect to it from this bastion host.

10. From the **Amazon EC2 console**, create an EC2 instance in one of the public subnets of the *Lab VPC*. It must meet the following criteria:
 - Name: **Bastion Host**
 - **Amazon Machine Image (AMI)**: *Amazon Linux 2023 AMI*
 - **Instance type**: *t2.micro*
 - Uses the **vockey** key pair
 - **Auto-assign Public IP**: This setting should be enabled
 - Only allows the following traffic:
 - **Type**: *SSH*
 - **Port**: **22**
 - **Source**: Your IP address

Module 9 Challenge Lab - Cre...c92520a2061448152182451w7523...Launch an instance | EC2 | us-east-1

https://us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances:

awsServicesSearch[Alt+S]N. Virginiavoclabs/user2752985=103488117@student.swin.edu.au @ 7523-2481...

Name and tagsInfo

Name

Bastion Host

Add additional tags

▼ Application and OS Images (Amazon Machine Image)Info

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

Search our full catalog including 1000s of application and OS images

Recents

My AMIs

Quick Start

Amazon Linux

aws

macOS

Mac

Ubuntu

ubuntu

Windows

Microsoft

Red Hat

Red Hat

SL

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 AMI

ami-05c13eab67c5d8861 (64-bit (x86)) / ami-0840becec4971bb87 (64-bit (Arm))

Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Description

Amazon Linux 2023 AMI 2023.2.20231030.1 x86_64 HVM kernel-6.1

Architecture

64-bit (x86)

AMI ID

ami-05c13eab67c5d8861

Verified provider

▼ Instance typeInfo

Instance type

t2.micro

Family: t2 1 vCPU 1 GiB Memory Current generation: true

On-Demand Windows base pricing: 0.0162 USD per Hour

On-Demand SUSE base pricing: 0.0116 USD per Hour

On-Demand RHEL base pricing: 0.0716 USD per Hour

On-Demand Linux base pricing: 0.0116 USD per Hour

Free tier eligible

All generations

Compare instance types

Additional costs apply for AMIs with pre-installed software

▼ Summary

Number of instancesInfo

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.2.2...read more

ami-05c13eab67c5d8861

Virtual server type (instance type)

t2.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Free tier:

In your first year includes 750 hours of t2.micro (or t3.micro in the Regions in which t2.micro is unavailable) instance usage on free tier AMIs per month, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet.

Cancel

Launch instance

Review commands

CloudShellFeedback

© 2023, Amazon Web Services, Inc. or its affiliates. PrivacyTermsCookie preferences

Module 9 Challenge Lab - Create a new EC2 instance

c92520a2061448152182451w7523

Launch an instance | EC2 | us-east-1

https://us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances:

aws

Services

Search

[Alt+S]

N. Virginia

voclabs/user2752985=103488117@student.swin.edu.au @ 7523-2481...

▼ Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

vockey

Create new key pair

▼ Network settings Info

VPC - required Info

vpc-053eb1d0788340786 (Lab VPC)

Subnet Info

subnet-086302573879b60f5 Public Subnet 1

Auto-assign public IP Info

Enable

Firewall (security groups) Info

Create security group

Select existing security group

Security group name - required

Bastion Host SG

Description - required Info

launch-wizard-1 created 2023-11-09T02:04:30.094Z

Inbound Security Group Rules

Security group rule 1 (TCP, 22, 116.96.47.180/32)

Type Info

ssh

Protocol Info

TCP

Port range Info

22

Source type Info

My IP

Name Info

116.96.47.180/32

Description - optional Info

e.g. SSH for admin desktop

Add security group rule

Advanced network configuration

▼ Summary

Number of instances Info

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.2.2...read more

Virtual server type (instance type)

t2.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year includes 750 hours of t2.micro (or t3.micro in the Regions in which t2.micro is unavailable) instance usage on free tier AMIs per month, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet.

Cancel

Launch instance

Review commands

CloudShell

Feedback

© 2023, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

Task 4: Creating a launch template

During the lab setup, an Amazon Machine Image (AMI) was created from the *CafeWebAppServer* instance. In this task, you will create a launch template by using this AMI.

11. Create a launch template by using the AMI that was created during lab setup. It must meet the following criteria.
 - **AMI:** *Cafe WebServer Image*
Tip: To locate the AMI, go to the **AMI** dropdown menu and enter: *Cafe*
 - **Instance type:** *t2.micro*
Tip: To locate the instance type, go to the **Instance Type** dropdown menu and enter: *t2*
 - **Key pair (login):** Uses a *new key pair* **Tip:** Create a new key pair and select it. Make sure that you download the key pair to your local computer.
 - **Security groups:** *CafeSG*
Tip: To locate the security group, go to the **Security groups** dropdown menu and enter: *CafeSG*
 - **Resource tags:**
 - **Key:** *Name*
 - **Value:** *webserver*
 - **Resource types:** *Instances*
 - **IAM Instance Profile:** *CafeRole*
Tip: Look in **Advanced Details** for this setting.

Create key pair

Key pair name
Key pairs allow you to connect to your instance securely.
LTDA
The name can include upto 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ RSA
RSA encrypted private and public key pair

☐ ED25519
ED25519 encrypted private and public key pair

Private key file format

☐ .pem
For use with OpenSSH

☒ .ppk
For use with PuTTY

⚠ When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance. [Learn more](#)

Cancel Create key pair

(I later converted the .ppk file to .pem)

Module 9 Challenge Lab - Create AMI

c92520a206144815218245t1w7523

Create launch template | EC2

https://us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateTemplate:

aws

Services

Search

[Alt+S]

N. Virginia

voclabs/user2752985=103488117@student.swin.edu.au @ 7523-2481...

▼ Application and OS Images (Amazon Machine Image) Info

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

Search our full catalog including 1000s of application and OS images

RecentsMy AMIsQuick Start

☐ Don't include in launch template

☒ Owned by me

☐ Shared with me

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Cafe WebServer Image
ami-0d0d3e9686bd1efae
2023-11-09T01:44:25.000Z Virtualization: hvm ENA enabled: true Root device type: ebs

Description

-

ArchitectureAMI ID

x86_64ami-0d0d3e9686bd1efae

▼ Instance type Info Advanced

Instance type

t2.micro

Family: t2 1 vCPU 1 GiB Memory Current generation: true
On-Demand Windows base pricing: 0.0162 USD per Hour
On-Demand SUSE base pricing: 0.0116 USD per Hour
On-Demand RHEL base pricing: 0.0716 USD per Hour
On-Demand Linux base pricing: 0.0116 USD per Hour

Free tier eligible

☐ All generations

Compare instance types

Additional costs apply for AMIs with pre-installed software

▼ Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name

LTDA

Create new key pair

▼ Summary

Software Image (AMI)

Cafe WebServer Image
ami-0d0d3e9686bd1efae

Virtual server type (instance type)

t2.micro

Firewall (security group)

c92520a206144815218245t1w752324815427-CafeSG-1X2REZQQUNAP5

Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year includes 750 hours of t2.micro (or t3.micro in the Regions in which t2.micro is unavailable) instance usage on free tier AMIs per month, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet.

Cancel

Create launch template

CloudShell Feedback

© 2023, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

Module 9 Challenge Lab - Create EC2 instance

https://us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateTemplate:

Services Search [Alt+S]

N. Virginia

voclabs/user2752985=103488117@student.swin.edu.au @ 7523-2481...

Network settings

Subnet Info

Don't include in launch template

Create new subnet

When you specify a subnet, a network interface is automatically added to your template.

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Select existing security group Create security group

Security groups Info

Select security groups

c92520a206144815218245t1w752324815427-CafeSG-1X2REZQQUNAP5

sg-06c27acf4cdd13f98

Compare security group rules

Advanced network configuration

Storage (volumes)

EBS Volumes Hide details

Volume 1 (AMI Root) (8 GiB, EBS, General purpose SSD (gp2))

AMI Volumes are not included in the template unless modified

Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage

Add new volume

Resource tags

Key Info Value Info Resource types Info

Name webserver

Select resource type

Instances

Add new tag

You can add up to 49 more tags.

Summary

Software Image (AMI)

Cafe WebServer Image

ami-0d0d3e9686bd1efae

Virtual server type (instance type)

t2.micro

Firewall (security group)

c92520a206144815218245t1w752324815427-CafeSG-1X2REZQQUNAP5

Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year includes 750 hours of t2.micro (or t3.micro in the Regions in which t2.micro is unavailable) instance usage on free tier AMIs per month, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet.

Cancel Create launch template

CloudShell Feedback

© 2023, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

Module 9 Challenge Lab - Create EC2 instance

https://us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateTemplate:

Services Search [Alt+S]

N. Virginia

voclabs/user2752985=103488117@student.swin.edu.au @ 7523-2481...

Advanced details

IAM instance profile Info

CafeRole

arn:aws:iam::752324815427:instance-profile/CafeRole

Create new IAM profile

Summary

Software Image (AMI)

Cafe WebServer Image

ami-0d0d3e9686bd1efae

Task 5: Creating an Auto Scaling group

Now that the launch template is defined, you will create an Auto Scaling group for the instances. In this task, do *not* create a load balancer when you create the Auto Scaling group. (You will create a load balancer in the next task.)

12. Create a new Auto Scaling Group that meets the following criteria:
 - **Launch template:** Uses the launch template that you created in the previous task
 - **VPC:** Uses the VPC that was configured for this lab
 - **Subnets:** Uses **Private Subnet 1** and **Private Subnet 2**
 - Skips *all* the advanced options
 - Has a **Group size** configured as:
 - **Desired capacity:** **2**
 - **Minimum capacity:** **2**
 - **Maximum capacity:** **6**
 - Enables the **Target tracking scaling policy** configured as:
 - **Metric type:** *Average CPU utilization*
 - **Target Value:** **25**
 - **Instances need:** **60**

Module 9 Challenge Lab - Create Auto Scaling group

https://us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateAutoScalingGroup

ServicesSearch[Alt+S]

N. Virginiavoclabs/user2752985=103488117@student.swin.edu.au @ 7523-2481...

EC2 > Auto Scaling groups > Create Auto Scaling group

Step 1

Choose launch template

Step 2

Choose instance launch options

Step 3 - optional

Configure advanced options

Step 4 - optional

Configure group size and scaling policies

Step 5 - optional

Add notifications

Step 6 - optional

Add tags

Step 7

Review

Choose launch template

Specify a launch template that contains settings common to all EC2 instances that are launched by this Auto Scaling group.

Name

Auto Scaling group name

Enter a name to identify the group.

MyASG

Must be unique to this account in the current Region and no more than 255 characters.

Launch template

For accounts created after May 31, 2023, the EC2 console only supports creating Auto Scaling groups with launch templates. Creating Auto Scaling groups with launch configurations is not recommended but still available via the CLI and API until December 31, 2023.

Launch template

Choose a launch template that contains the instance-level settings, such as the Amazon Machine Image (AMI), instance type, key pair, and security groups.

MyTemplate

Create a launch template

Version

Default (1)

Create a launch template version

Description	Launch template	Instance type
-	MyTemplate	t2.micro
	lt-0c64ddb94aa56b4f4	
AMI ID	Security groups	Request Spot Instances
ami-0d0d3e9686bd1efae	-	No
Key pair name	Security group IDs	
LTDA	sg-06c27acf4cdd13f98	

Additional details

Storage (volumes)	Date created
-	Thu Nov 09 2023 09:16:30 GMT+0700 (Indochina Time)

Cancel

Next

CloudShellFeedback

© 2023, Amazon Web Services, Inc. or its affiliates. PrivacyTermsCookie preferences

Module 9 Challenge Lab - Create Auto Scaling group

Create Auto Scaling group | EC2

https://us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateAutoScalingGroup

aws Services Search [Alt+S] N. Virginia voclabs/user2752985=103488117@student.swin.edu.au @ 7523-2481...

EC2 > Auto Scaling groups > Create Auto Scaling group

Step 1
[Choose launch template](#)

Step 2
Choose instance launch options

Step 3 - optional
[Configure advanced options](#)

Step 4 - optional
[Configure group size and scaling policies](#)

Step 5 - optional
[Add notifications](#)

Step 6 - optional
[Add tags](#)

Step 7
[Review](#)

Choose instance launch options [Info](#)

Choose the VPC network environment that your instances are launched into, and customize the instance types and purchase options.

Instance type requirements [Info](#)

Override launch template

You can keep the same instance attributes or instance type from your launch template, or you can choose to override the launch template by specifying different instance attributes or manually adding instance types.

Launch template	Version	Description
MyTemplate lt-0c64ddb94aa56b4f4	Default	-

Instance type
t2.micro

Network [Info](#)

For most applications, you can use multiple Availability Zones and let EC2 Auto Scaling balance your instances across the zones. The default VPC and default subnets are suitable for getting started quickly.

VPC
Choose the VPC that defines the virtual network for your Auto Scaling group.

vpc-053eb1d0788340786 (Lab VPC)
10.0.0.0/16

Create a VPC

Availability Zones and subnets
Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.

Select Availability Zones and subnets

us-east-1a | subnet-0371035b526ef00be (Private Subnet 1)
10.0.2.0/24

us-east-1b | subnet-035da0feecd9fa69d (Private Subnet 2)
10.0.3.0/24

Create a subnet

Cancel

Skip to review

Previous

Next

CloudShell Feedback

© 2023, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

The screenshot shows the AWS Management Console interface for creating an Auto Scaling group. The browser address bar indicates the URL is `https://us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateAutoScalingGroup`. The console header shows the user is logged in as `voclabs/user2752985=103488117@student.swin.edu.au` in the `N. Virginia` region.

The left sidebar contains a navigation menu with the following steps:

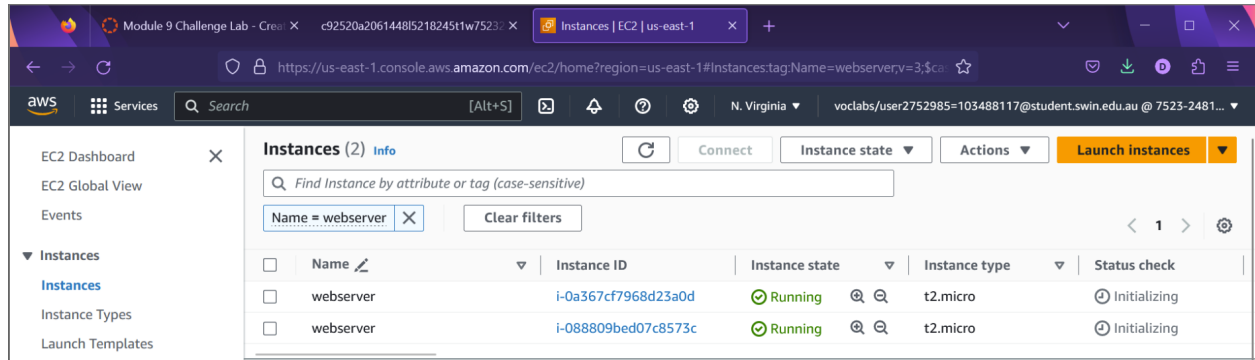
- Step 1: [Choose launch template](#)
- Step 2: [Choose instance launch options](#)
- Step 3 - optional: [Configure advanced options](#)
- Step 4 - optional: **Configure group size and scaling policies**
- Step 5 - optional: [Add notifications](#)
- Step 6 - optional: [Add tags](#)
- Step 7: [Review](#)

The main content area is titled **Configure group size and scaling policies - optional**. It includes the following sections:

- Group size - optional**: A section for specifying the size of the Auto Scaling group. It includes three input fields:
 - Desired capacity**: Set to 2.
 - Minimum capacity**: Set to 2.
 - Maximum capacity**: Set to 6.
- Scaling policies - optional**: A section for choosing whether to use a scaling policy. It includes:
 - Target tracking scaling policy**: Selected with a radio button. Description: "Choose a desired outcome and leave it to the scaling policy to add and remove capacity as needed to achieve that outcome."
 - None**: Unselected with a radio button.
 - Scaling policy name**: A text input field containing "Target Tracking Policy".
 - Metric type**: A dropdown menu set to "Average CPU utilization".
 - Target value**: A text input field containing "25".
 - Instance warmup**: A text input field containing "60" with the unit "seconds".
 - ☐ **Disable scale in to create only a scale-out policy**.

The footer of the console shows "CloudShell", "Feedback", and copyright information: "© 2023, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences".

13. To verify that you created the Auto Scaling group correctly, go to the **Amazon EC2 console**. You should have two instances, both with the name that you configured as *resource tags* in the previous task.



Task 6: Creating a load balancer

Now that you web application server instances are deployed in private subnets, you need a way for the outside world to connect to them. In this task, you will create a load balancer to distribute traffic across your private instances.

14. Create an HTTP Application Load Balancer that meets the following criteria:
 - **VPC:** Uses the VPC configured for this lab
 - **Subnets:** Uses the two *public subnets*
 - Skips the HTTPS security configuration settings
 - **Security group:** Creates a *new security group* that allows HTTP traffic from anywhere
 - **Target group:** Creates a *new target group*
 - Skips registering targets

Note: *Wait* until the load balancer is active.

Module 9 Challenge Lab - Create application load balancer

EC2 | us-east-1

https://us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateSecurityGroup

Services Search [Alt+S]

N. Virginia

voclabs/user2752985=103488117@student.swin.edu.au @ 7523-2481...

EC2 > Security Groups > Create security group

Create security group [Info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name [Info](#)
ALB SG
Name cannot be edited after creation.

Description [Info](#)
Allow HTTP traffic from anywhere

VPC [Info](#)
vpc-053eb1d0788340786 (Lab VPC)

Inbound rules [Info](#)

Type	Protocol	Port range	Source	Description - optional	
HTTP	TCP	80	An... 0.0.0.0/0		Delete
Add rule					

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Outbound rules [Info](#)

Type	Protocol	Port range	Destination	Description - optional	
All traffic	All	All	Cu... 0.0.0.0/0		Delete
Add rule					

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

CloudShell Feedback

© 2023, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

Module 9 Challenge Labc92520a2061448f521824511Create application loadStep 1 Create target groupEC2 | us-east-1

https://us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateTargetGroup:protocol=HTTP;vpc-

awsServicesSearch[Alt+S]

N. Virginiavoclabs/user2752985=103488117@student.swin.edu.au @ 7523-2481...

Register targets

Basic configuration

Settings in this section can't be changed after the target group is created.

Choose a target type

☒ Instances

- Supports load balancing to instances within a specific VPC.
- Facilitates the use of [Amazon EC2 Auto Scaling](#) to manage and scale your EC2 capacity.

☐ IP addresses

- Supports load balancing to VPC and on-premises resources.
- Facilitates routing to multiple IP addresses and network interfaces on the same instance.
- Offers flexibility with microservice based architectures, simplifying inter-application communication.
- Supports IPv6 targets, enabling end-to-end IPv6 communication, and IPv4-to-IPv6 NAT.

☐ Lambda function

- Facilitates routing to a single Lambda function.
- Accessible to Application Load Balancers only.

☐ Application Load Balancer

- Offers the flexibility for a Network Load Balancer to accept and route TCP requests within a specific VPC.
- Facilitates using static IP addresses and PrivateLink with an Application Load Balancer.

Target group name

TG

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Protocol : Port

HTTP80

1-65535

IP address type

Only targets with the indicated IP address type can be registered to this target group.

☒ IPv4

- Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.

☐ IPv6

- Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). [Learn more](#)

VPC

Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

Lab VPC
vpc-053eb1d0788340786
IPv4: 10.0.0.0/16

Protocol version

☒ HTTP1

- Send requests to targets using HTTP/1.1. Supported when the request protocol is HTTP/1.1 or HTTP/2.

CloudShellFeedback

© 2023, Amazon Web Services, Inc. or its affiliates. PrivacyTermsCookie preferences

Module 9 Challenge Labc92520a2061448f5218245t1Create application loadTarget group details | EC2 | us-east-1

←→↻https://us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateALBWizard:☆📄📌🔍

awsServices🔍 Search[Alt+S]📧🔔🔗⚙️N. Virginia▼voclabs/user2752985=103488117@student.swin.edu.au @ 7523-2481...

☰

Basic configuration

Load balancer name

Name must be unique within your AWS account and can't be changed after the load balancer is created.

ALB

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme [Info](#)

Scheme can't be changed after the load balancer is created.

☒ Internet-facing

An internet-facing load balancer routes requests from clients over the internet to targets. Requires a public subnet. [Learn more](#)

☐ Internal

An internal load balancer routes requests from clients to targets using private IP addresses.

IP address type [Info](#)

Select the type of IP addresses that your subnets use.

☒ IPv4

Recommended for internal load balancers.

☐ Dualstack

Includes IPv4 and IPv6 addresses.

Network mapping [Info](#)

The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC [Info](#)

Select the virtual private cloud (VPC) for your targets or you can [create a new VPC](#). Only VPCs with an internet gateway are enabled for selection. The selected VPC can't be changed after the load balancer is created. To confirm the VPC for your targets, view your [target groups](#).

Lab VPC

vpc-053eb1d0788340786

IPv4: 10.0.0.0/16

↻

Mappings [Info](#)

Select at least two Availability Zones and one subnet per zone. The load balancer routes traffic to targets in these Availability Zones only. Availability Zones that are not supported by the load balancer or the VPC are not available for selection.

☒ us-east-1a (use1-az1)

Subnet

subnet-086302573879b60f5Public Subnet 1 ▼

IPv4 address

Assigned by AWS

☒ us-east-1b (use1-az2)

Subnet

subnet-0135bfa49a0880184Public Subnet 2 ▼

IPv4 address

Assigned by AWS

☐ us-east-1c (use1-az4)

☐ us-east-1d (use1-az6)

CloudShellFeedback

© 2023, Amazon Web Services, Inc. or its affiliates. [Privacy](#) [Terms](#) [Cookie preferences](#)

Name: Trac Duc Anh Luong - ID: 103488117

The screenshot shows the AWS Management Console for the 'Create application load balancer' wizard. The browser tabs include 'Module 9 Challenge Lab', 'c92520a2061448f52182451...', 'Create application load balancer', 'Target group details | EC2', and 'EC2 | us-east-1'. The URL is 'https://us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateALBWizard:'. The 'Security groups' section shows a dropdown with 'Select up to 5 security groups' and a button 'ALB SG' with details 'sg-0208c23e3f1ed2d16 VPC: vpc-053eb1d0788340786'. The 'Listeners and routing' section shows a listener for 'HTTP:80' with a default action of 'Forward to TG' (Target type: Instance, IPv4). There is a 'Remove' button for the listener and an 'Add listener' button at the bottom.

15. Modify the Auto Scaling group that you created in the previous task by adding this new load balancer.

Hint: Add the *target group* you created in the Load Balancer configuration.

The screenshot shows the AWS Management Console for the 'Edit Auto Scaling group' page. The browser tabs include 'Module 9 Challenge Lab', 'c92520a2061448f52182451...', 'Edit Auto Scaling group', 'Target group details | EC2', and 'EC2 | us-east-1'. The URL is 'https://us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#EditAutoScalingGroup:id=MyASG'. The left sidebar shows the navigation menu with 'Instances' and 'Images' expanded. The 'Load balancing - optional' section shows 'Application, Network or Gateway Load Balancer target groups' selected, with a dropdown showing 'Select target groups' and a button 'TG | HTTP' (Application Load Balancer: ALB). There is also a 'Classic Load Balancers' checkbox and an 'Add a new load balancer' button.

Name: Trac Duc Anh Luong - ID: 103488117

Sofia has now created and configured the resources that the web application needs to be highly available and scalable. However, Sofia knows that she still has more work to do. To complete the process of updating the application architecture, Sofia must test the café's web application to make sure that it performs as expected.

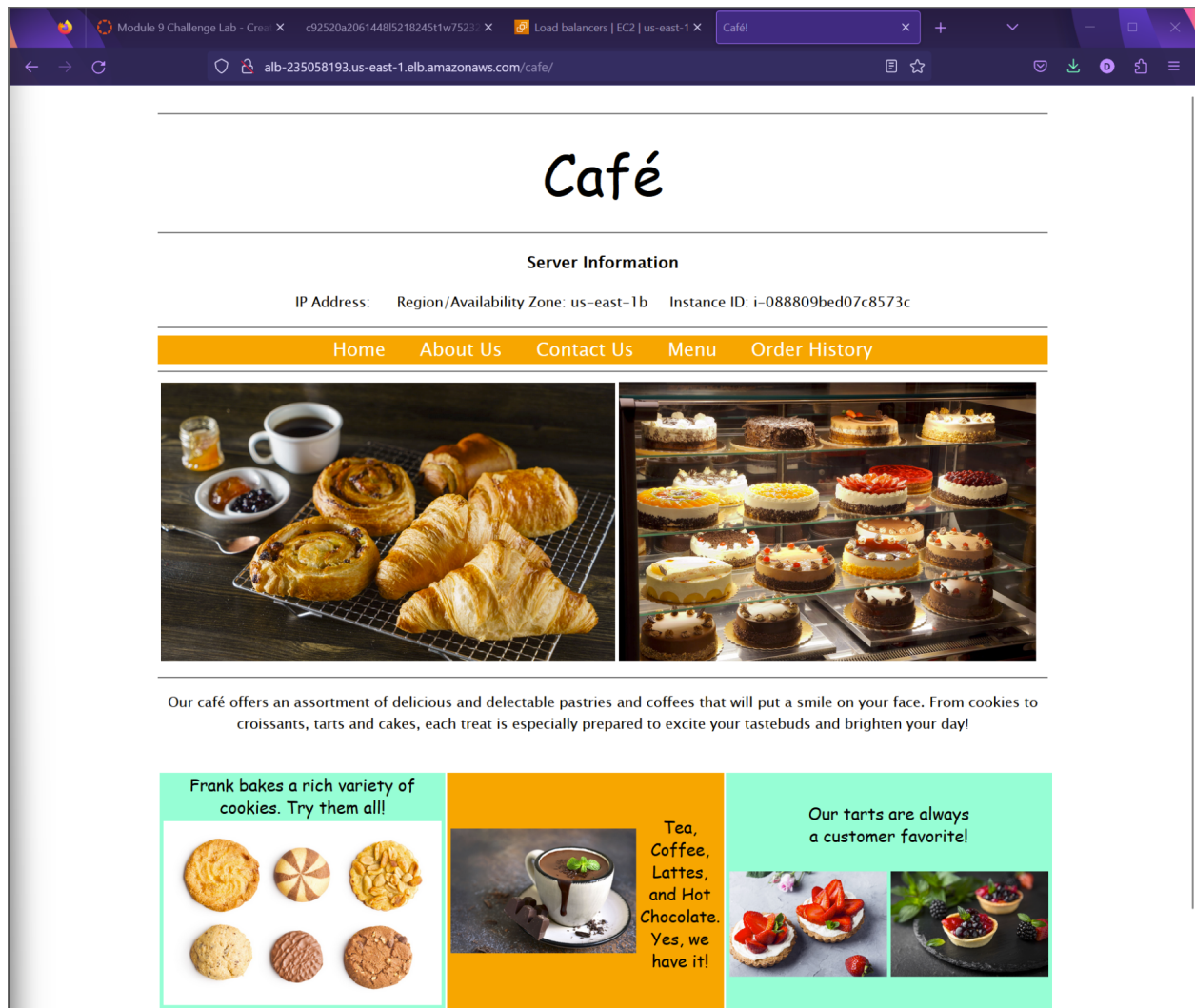
In the next tasks, you will continue working in the role of Sofia and test whether the café web application automatically scales under load.

Task 7: Testing the web application

In this task, you will test the café web application.

16. To test the café web application, visit the Domain Name System (DNS) name of your load balancer and append `/cafe` to the URL.

The café application should load.

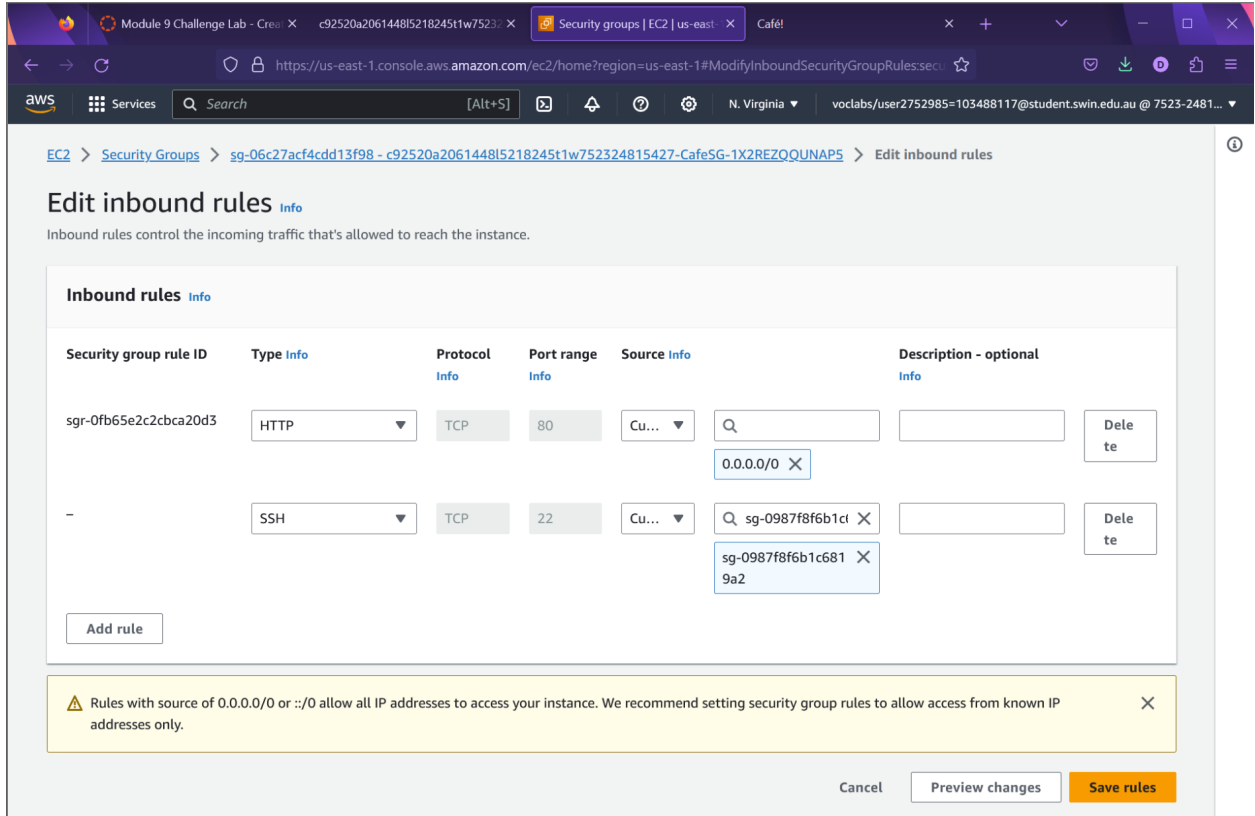


Task 8: Testing automatic scaling under load

In this task, you will test whether the café application *scales out* automatically.

17. By using *Secure Shell (SSH) passthrough through the bastion host instance*, use SSH to connect to one of the running web server instances.

Tip: You will need to modify the *CafeSG* security group to allow SSH traffic over port 22 from the bastion host.



The screenshot shows the AWS Management Console interface for editing inbound rules on a security group. The breadcrumb navigation indicates the path: EC2 > Security Groups > sg-06c27acf4cdd13f98 - c92520a206144815218245t1w752324815427-CafeSG-1X2REZQQUNAP5 > Edit inbound rules. The page title is 'Edit inbound rules' with an 'Info' link. Below the title, a note states: 'Inbound rules control the incoming traffic that's allowed to reach the instance.'

The 'Inbound rules' section contains a table with the following columns: Security group rule ID, Type, Protocol, Port range, Source, and Description - optional. There are two rules listed:

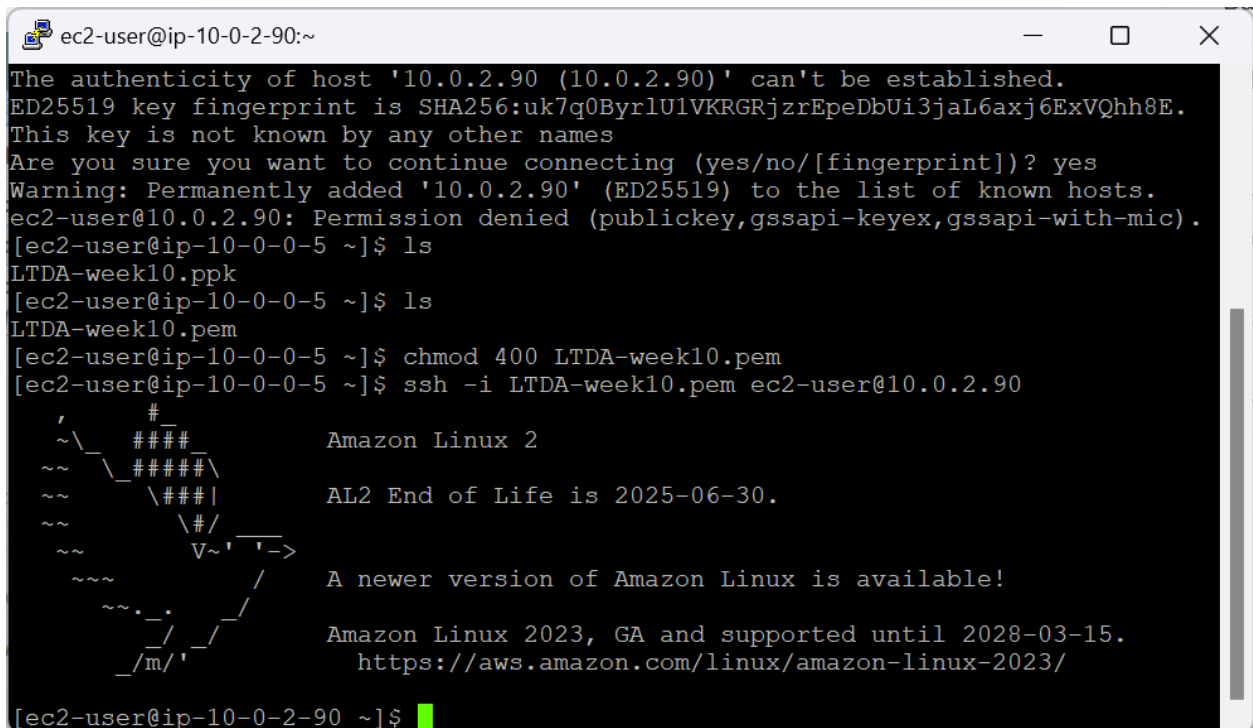
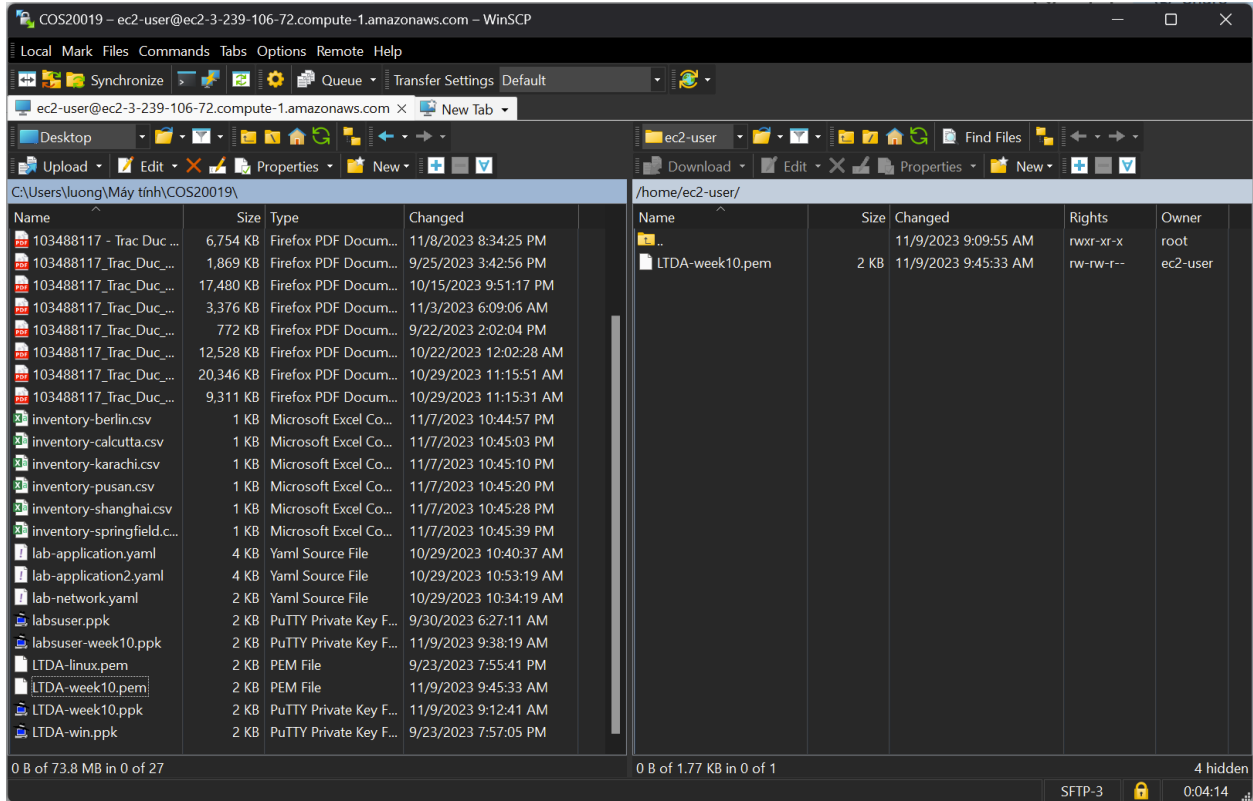
Security group rule ID	Type	Protocol	Port range	Source	Description - optional
sgr-0fb65e2c2cbca20d3	HTTP	TCP	80	Cu...	
-	SSH	TCP	22	Cu...	

The SSH rule is selected, and its 'Source' field is being edited. The current source is '0.0.0.0/0', which is being replaced with 'sg-0987f8f6b1c6819a2'. A search bar is visible above the source field, and a 'Delete' button is next to the rule. An 'Add rule' button is at the bottom left of the rules list.

A yellow warning banner at the bottom of the console states: 'Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.'

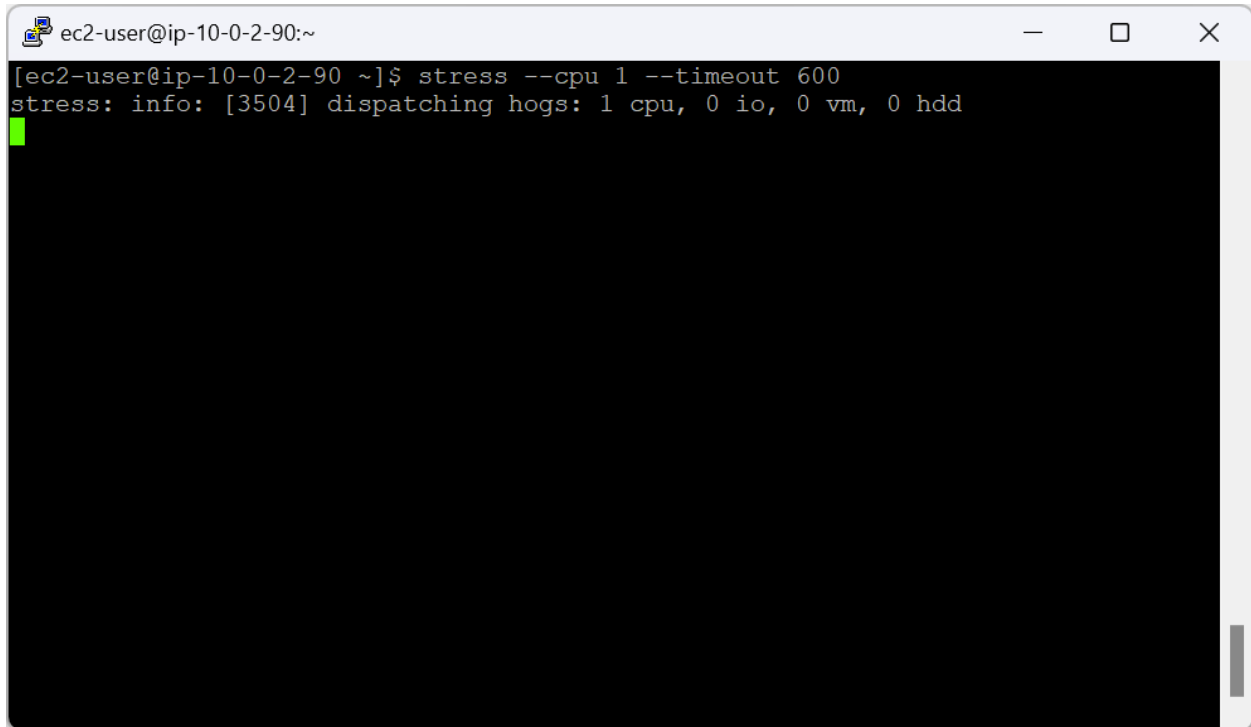
At the bottom right, there are three buttons: 'Cancel', 'Preview changes', and 'Save rules'.

Name: Trac Duc Anh Luong - ID: 103488117



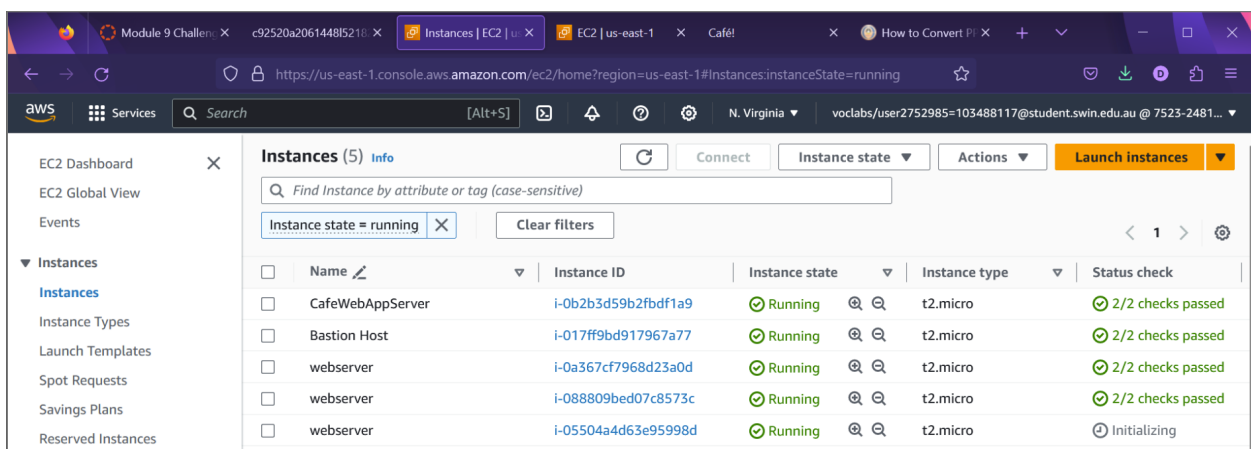
18. From the web server instance, use the following commands to start a stress test. This test increases the load on the web server CPU:

```
sudo yum install https://dl.fedoraproject.org/pub/epel/epel-release-latest-7.noarch.rpm
sudo yum install stress -y
stress --cpu 1 --timeout 600
```



A terminal window titled 'ec2-user@ip-10-0-2-90:~' showing the execution of the stress test. The command 'stress --cpu 1 --timeout 600' is entered, and the output is 'stress: info: [3504] dispatching hogs: 1 cpu, 0 io, 0 vm, 0 hdd'. The terminal background is black with white text.

19. Verify that the Auto Scaling group deploys new instances.
- Continue to observe the Amazon EC2 console.
 - During the test, you should observe that more web server instances are deployed.



A screenshot of the Amazon EC2 console. The 'Instances' tab is selected, showing a list of 5 instances. The instances are: CafeWebAppServer, Bastion Host, and three webserver instances. All instances are in the 'Running' state and have '2/2 checks passed'.

Name	Instance ID	Instance state	Instance type	Status check
CafeWebAppServer	i-0b2b3d59b2fbd1a9	Running	t2.micro	2/2 checks passed
Bastion Host	i-017ff9bd917967a77	Running	t2.micro	2/2 checks passed
webserver	i-0a367cf7968d23a0d	Running	t2.micro	2/2 checks passed
webserver	i-088809bed07c8573c	Running	t2.micro	2/2 checks passed
webserver	i-05504a4d63e95998d	Running	t2.micro	Initializing

Update from the café

After Sofia finishes testing the café web application's performance, she tells her parents about the changes—when traffic increases, the café application now successfully scales out. Frank and Martha are impressed that Sofia implemented a highly available and scalable architecture for the application.

Meanwhile, the entire café team is excited! They are busy preparing for the increased volume of orders that they anticipate from being featured on the TV show. As they work on their preparations, they know that they can rely on automatic scaling to help them take orders and delight new customers.

Submitting your work

20. At the top of these instructions, choose Submit to record your progress and when prompted, choose **Yes**.
21. If the results don't display after a couple of minutes, return to the top of these instructions and choose Grades
Tip: You can submit your work multiple times. After you change your work, choose **Submit** again. Your last submission is what will be recorded for this lab.
22. To find detailed feedback on your work, choose Details followed by **View Submission Report**.

The screenshot shows the AWS Management Console interface for a lab submission. The sidebar on the left contains navigation links: Courses, Grades (with a notification badge), Calendar, Inbox, History, and Help. The main content area has a top navigation bar with buttons: Submit, Details (selected), AWS, Start Lab, End Lab, 1:41, Instructions, Grades, and Actions. Below this is a tabbed interface with Files, README (checked), Terminal (checked), and Source. The Terminal window displays the submission report, including the execution time (Wed Nov 8 18:52:28 PST 2023), answers to six questions, and testing results for seven tasks. A table on the right summarizes the scores.

Task	Score
Total score	36/36
[Answer 01]	1/1
[Answer 02]	1/1
[Answer 03]	1/1
[Answer 04]	1/1
[Answer 05]	1/1
[Answer 06]	1/1
[Task 2] NAT Gateway exists	5/5
[Task 3] Bastion Host exists	5/5
[Task 4] Launch Template	5/5
[Task 5] Auto Scaling Group	5/5
[Task 6] Load Balancer	5/5
[Task 7] Testing the web application	5/5